

CS610 Short Notes Mid term

Lec 1 to22

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Lecture No. 1

Network

System for connecting computers using a single transmission technology

Internet

Set of networks connected by routers that are configured to pass traffic among any computers attached to any network in the set.

CLASSIFICATION OF NETWORKS

Computer networks are classified by four factors which are as follow:

- BY SIZE:
- BY CONNECTIVITY:
- BY MEDIUM:
- BY MOBILITY:

1) BY SIZE:

According to their size there are two classifications of networks.

- I. Local Area Network. (LAN)
- II. Wide Area Network (WAN)

Local Area Network. (LAN)

In LAN network occupies the smaller area like a room a floor or a building.

Wide Area Network (WAN)

In WAN, network occupies larger areas like cities & countries. Internet is a Wide Area Network.

2) BY CONNECTIVITY:

Networks are also classified by connectivity in which two topologies are discussed.

- I. Point-to-Point

II. Broadcast

POINT-TO-POINT:

Point-to-Point topologies there are two topologies.

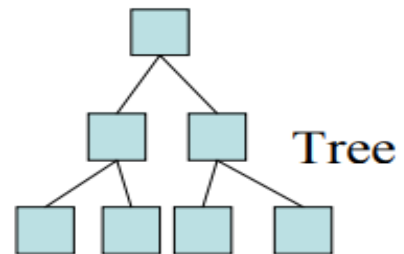
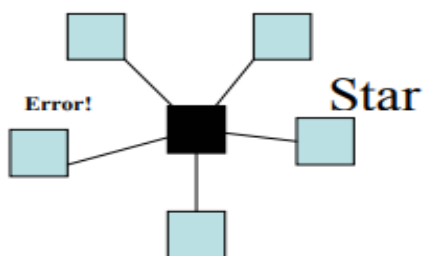
- 1) STAR topology
- 2) TREE topology

STAR topology

In star topology each computer is connected to a central hub. The communication takes place through the hub.

TREE topology

In Tree topology all computers are connected to each other in such a way that they make a tree.



BROADCAST:

In broadcast topology there are further two categories

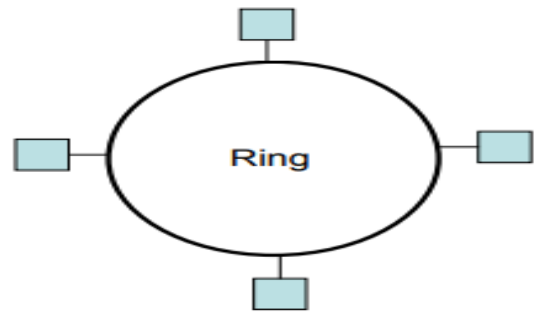
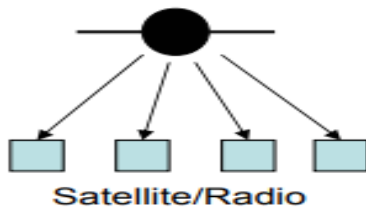
- 1) SATELLITE\RADIO
- 2) RING TOPOLOGY

SATELLITE\RADIO

In a satellite or radio topology all computers are connected to each other via satellite or radio wave.

RING TOPOLOGY

Satellite and Ring topologies: In a ring topology each computer is connected to other thorough a ring.



3) BY MEDIUM:

The classification of networks is also based on the Medium of transmission. Following are the mediums of transmission:

- Copper wire
- Co-axial cable
- Optical fiber
- Radio waves

4) BY MOBILITY:

The networks are also classified according to their mobility. In this respect there are two types of networks.

- Fixed networks
- Mobile networks

NETWORKS IN DAILY LIFE:

The major use of networks is in business side. Networks are used for advertising, production, shipping, planning, billing and accounting purposes. In fact now there is an entire industry that develops networking equipment.

CS610 Lecture-2

Resource Sharing

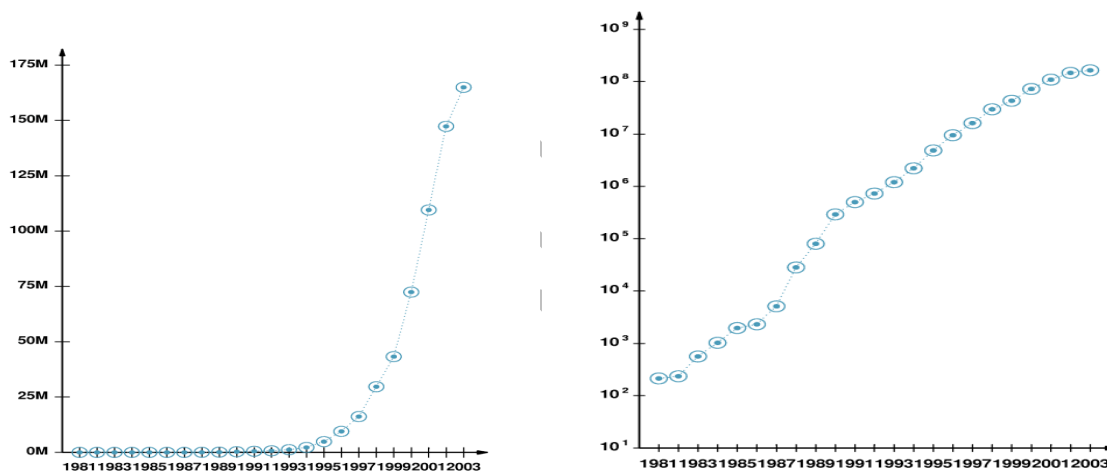
Sharing a peripheral device such as a printer or a disk.

Sharing a large-scale computational power, because

- Computers were extremely expensive
- The government budgets were not sufficient to provide computers for all scientists and engineers.

Growth of the Internet

U.S Department of Defence Advanced Research Projects Agency (ARPA) started ARPANET project. The early internet emerged in 1970s Has grown drastically since then.



Probing The Internet

Now an automated tool is required that tests to see whether the given computer is online or not. For this purpose the first tool is the 'PING program'

```
PING sears.com: 56 data bytes
64 bytes from 32.97.168.129: icmp_seq=0. time=49. ms
64 bytes from 32.97.168.129: icmp_seq=1. time=50. ms
64 bytes from 32.97.168.129: icmp_seq=2. time=48. ms
64 bytes from 32.97.168.129: icmp_seq=3. time=50. ms
64 bytes from 32.97.168.129: icmp_seq=4. time=48. ms
----sears.com PING Statistics----
5 packets transmitted, 5 packets received, 0% packet loss
round-trip (ms)  min/avg/max = 48/49/50
```

PROBLEM WITH 'PING':

- "Ping" cannot tell the reason of the problem
- Some networks or computers reject the ping packets
- "Traceroute"

Tracing a Route:

There is another probing tool i-e Trace Route. To get more detail it is used.

```
traceroute to DANDELION-PATCH.MIT.EDU (18.181.0.31), 40 byte packets
 1  cisco1 (128.10.2.250)  2 ms  1 ms  2 ms
 2  cisco-tel-252.tcom.purdue.edu (128.210.252.22)  2 ms  1 ms  1 ms
 3  abilene.tcom.purdue.edu (192.5.40.10)  6 ms  8 ms  7 ms
 4  clev-ipls.abilene.ucaid.edu (198.32.8.26)  14 ms  14 ms  12 ms
 5  nycm-clev.abilene.ucaid.edu (198.32.8.30)  24 ms  27 ms  24 ms
 6  192.5.89.45 (192.5.89.45)  31 ms  34 ms  35 ms
 7  192.5.89.10 (192.5.89.10)  33 ms  33 ms  33 ms
 8  NW12-RTR-FDDI.MIT.EDU (18.168.0.16)  59 ms  34 ms  33 ms
 9  DANDELION-PATCH.MIT.EDU (18.181.0.31)  62 ms * 79 ms
```

CS610 Lecture-3

This lecher is skipping.

CS610 Lecture-4 PACKETS, FRAMES AND ERROR DETECTION

PACKETS

A packet is the unit of data routed between an origin and a destination on the internet or other packet-switched network or networks that ship data around in small packets.



WITH PACKETS:

Now if the file is broken into packets, other computers must only wait until packet (not entire file) has been sent.

$$\begin{aligned} &= \frac{1000 \text{ bytes} * 8 \text{ bits / byte}}{56 \times 10^3 \text{ bits / second}} \\ &= .143 \text{ seconds} \end{aligned}$$

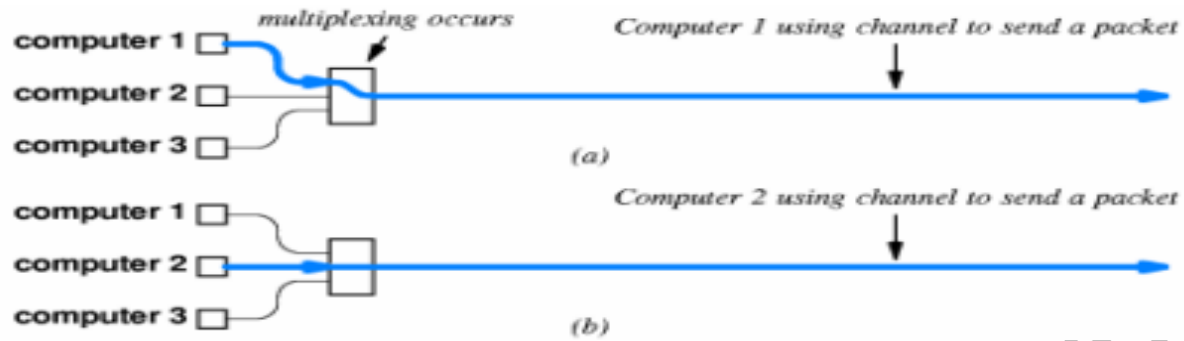
WITHOUT PACKETS:

A 5MB file transferred across network with 56Kpbs capacity will require 12 minutes. This means that all that computers will be forced to wait for 12 minutes before initiating other transfers.

$$\begin{aligned} &= \frac{5 \times 10^6 \text{ bytes} * 8 \text{ bits / byte}}{60 \text{ secs / minute} * 56 \times 10^3 \text{ bits / second}} \\ &= 11.9 \text{ minutes} \end{aligned}$$

PACKETS AND TDM:

Dividing data into small packets allow time division multiplexing. In TDM each packet leaves the source and is switched on the shared communication channel through a multiplexer. At the destination the packet is switched through a demultiplexer to the destination.



PACKETS AND FRAMES:

PACKETS:

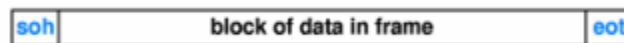
Packet is a generic term that refers to small block of data. Packet have different format. Each hardware uses different packet format.

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

FRAME FORMAT:

We need to define a standard format for data to indicate the beginning and end of the frame. Header and tail are used to frame the data.



FRAMING IN PRACTICE:

In practice there is a disadvantage of overhead. To avoid the no delay between two frames each frame sends an extra character between block of data. The framing in practice also has some transmission problems just like:

- Missing eot indicates sending computer crashed.
- Missing soh indicates receiving computer missed beginning of message.
- Bad frame is discarded.

Lecture No. 5 BYTE STUFFING

Data Stuffing

In general to distinguish between data being sent and control information such as frame delimiters network systems arrange for the sending side to change the data slightly before it is sent because systems usually insert data or bytes to change data for transmission, the technique is known as **Data Stuffing**.

There are two types of data stuffing:

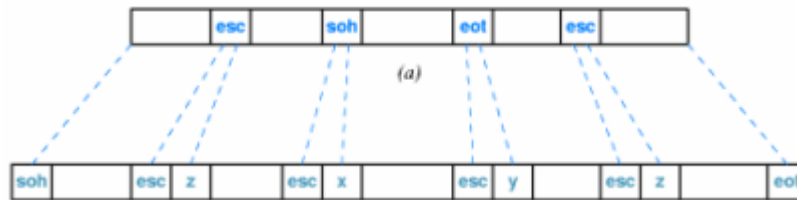
- Byte Stuffing
- Bit Stuffing

Byte stuffing

Byte stuffing refers stuffing with character oriented hardware and bit stuffing refers to bit oriented hardware.

Byte stuffing

Byte stuffing translates each reserved byte into two unreserved bytes. For example: it can use esc as prefix followed by x for soh, y for eot and z for eco.



TRANSMISSION ERRORS:

Transmission errors may occur due to different causes for example interference or power surges may destroy data during transmission. In result of which the bits are lost or the bit value may be changed.

ERROR DETECTION AND CORRECTION:

To detect and correct errors, frames include additional information, which is inserted by the sender and checked by the receiver. In this way incorrect data can be rejected. Also the incorrect data can be corrected and accepted.

PARITY CHECKING:

To detect the error there are different schemes in which parity checking is also commonly used. In parity checking, parity refers to the number of bits set to 1 in the data item.

There are two types:

- Even Parity
- Odd Parity

EVEN PARITY:

In an even parity the no. of 1's in data should be an even number.

ODD PARITY:

In an Odd parity the no. of bits should be an odd number.

PARITY BIT:

A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

For example, the binary value 10000000 has an odd parity. Therefore, a 0 would be added to keep the parity odd and a 1 would be added to give the value an even parity.

LIMITATIONS OF PARITY CHECKING:

LONG

Parity can only detect errors that change in odd number of bits for example the original data and parity is 10010001+1 (even parity) and the incorrect data is 10110011+1 (even parity). We see that even no. of bits have been changed due to noise so parity checking cannot detect this error. Parity usually is used to detect on bit error.

ALTERNATIVE ERROR DETECTION SCHEMES

In addition to parity checking alternative error detection mechanisms have been introduced. These mechanisms differ from each other by the following respects.

- The size of the additional information (transmission overhead)
- Computational complexity of the algorithm (computational overhead)
- The number of bits errors that can be detected (how well errors are detected)

CHECKSUM error detection

A checksum is an error-detection method in a the transmitter computes a numerical value according to the number of set or unset bits in a message and sends it along with each message frame. At the receiver end, the same checksum function (formula) is applied to the message frame to retrieve the numerical value.

CYCLIC REDUNDANCY CHECK (CRC):

A cyclic redundancy check (CRC) is an error-detecting code commonly used in digital networks and storage devices to detect accidental changes to digital data. Blocks of data entering these systems get a short check value attached, based on the remainder of a polynomial division of their contents.

CRC uses just two hardware components:

- Shift register
- Exclusive OR (XOR unit)

Lecture No. 6 SHIFT OPERATION

SHIFT OPERATION

This operation shifts all bits to the left one position.

For example



TYPES OF ERRORS in SHIFT OPERATION:

CRC can check the following errors better than check sums.

1. Vertical errors
2. Burst errors

VERTICAL ERRORS:

This type of error occurs due to a hardware failure. E.g. the second bit of every character will damage.

BURST ERRORS:

When a small set of bits changes near a specific location due to lightning or electric motor starting nearby etc. then these types of errors are called Burst errors.

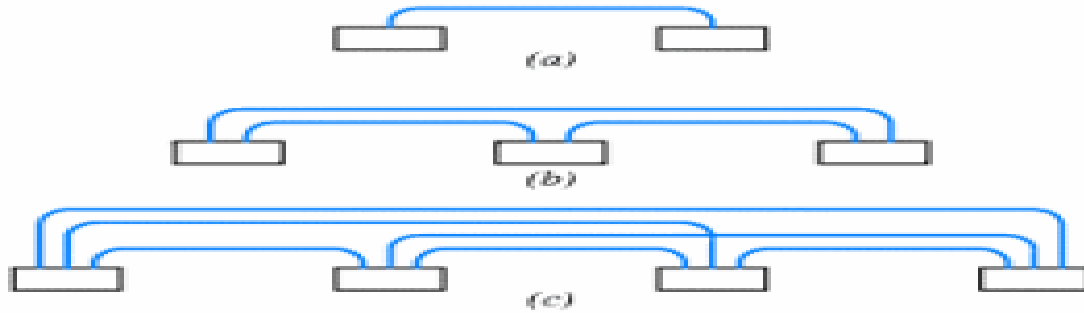
LAN TECHNOLOGY AND NETWORK TOPOLOGY

LONG

Most networks are local and are designed to share resources among multiple computers. Hardware technologies used for local networks allow multiple devices to connect with a shared network. In this shared medium the computers must take turns using the shared medium.

DIRECT POINT-TO-POINT COMMUNICATION:

Early networks used direct point-to-point communication. In such a mode of communication each communication channel connects exactly two computers. In this way it forms a mesh or point-to-point network,



ADVANTAGES:

Direct point-to-point communication has the following advantages:

- The connection type of individual connections can be different.
- Individual connections can choose a different frame format and error detection mechanism etc.
- It is easy to enforce security and privacy.

DISADVANTAGES:

Direct point-to-point communication has the following disadvantages:

- The no. of connections grow more rapidly than the no. of computers
- For 'n' computers connections = $(n^2 - n)/2$.
- Most computers use the same physical path.
- Direct point-to-point communication is expensive due to a no. of connections.
- Another disadvantage is that adding a new computer to the network requires N-1 new connections.

Lecture No. 7 GROWTH OF LAN TECHNOLOGY

GROWTH History OF LAN TECHNOLOGY

- The development of shared communication channels (LANs) started in 1960s and early 1970.
- The key idea behind was to reduce the number of connections by sharing connection among many computers Each LAN consists of a single shared medium.
- The computers take turns using the medium.
- First one computer uses the medium to send its data over the channel then second and son on. But sharing a single medium over long distances is efficient, due to the long delays.

- LAN technologies reduce cost by reducing no. of connections. But attached computers compete for use of shared connections.
- The local communication consists of LAN exclusively. But the long distance communication is point-to-point exclusively.

“LOCALITY OF REFERENCE” PRINCIPLE:

Principle of “Locality of Reference” helps predict computer communication patterns. There are two patterns given as follows:

- A) SPATIAL LOCALITY OF REFERENCE
- B) TEMPORAL LOCALITY OF REFERENCE

SPATIAL LOCALITY OF REFERENCE:

In this pattern computers are likely to communicate with other computers that are located nearby.

TEMPORAL LOCALITY OF REFERENCE:

In this pattern computers are likely to communicate with the same computers repeatedly. Thus LANs are effective because of spatial locality of reference. Temporal locality of reference may give insight into which computers should be on a LAN.

LAN TOPOLOGIES:

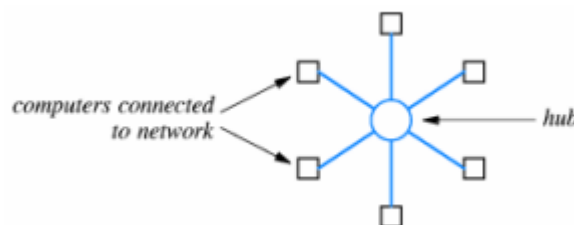
LONG

Network can be classified by shape. According to which there are three most popular topologies, which are given as follows;

- Star
- Ring
- Bus

STAR TOPOLOGY:

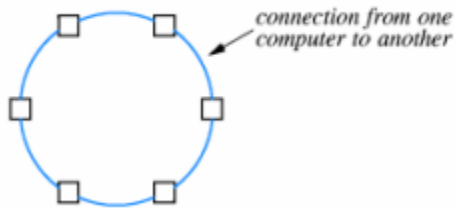
In this topology, all computers are attached to a central point, which is sometimes called



the “Hub”.

RING TOPOLOGY:

In this topology of network the computers are connected to each other in closed loop. In this network first computer passes data to the second and then second passes data to third and so on.



BUS TOPOLOGY:

In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

REASON FOR MULTIPLE TOPOLOGIES: LONG

Each topology has advantages and disadvantages, which are discussed below

1. **IN A RING:**

It is easy to coordinate access to other computers however entire network is disabled if a cable cut occurs.

2. **IN A STAR:**

On the other hand only once computer is affected when a cable cut occurs.

3. **IN A BUS:**

The network needs fewer wires than a star; however entire network is disabled when a cable cut occurs.

ETHERNET: LONG

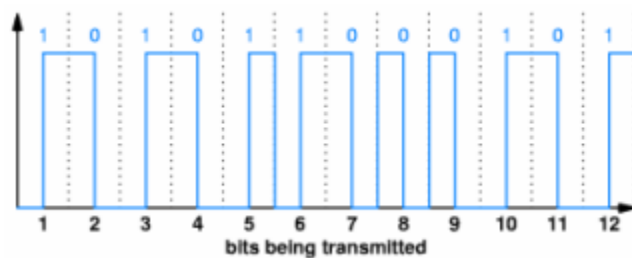
- Ethernet is a widely used LAN technology. It was invented at EXROX PARC (Palo Alto Research Center) in 1970s. Xerox, Intel and Digital defined it in a standard so it is also called DIX standard.
- The standard is now managed by IEEE in which 802.3 standard of IEEE defines formats, voltages of cable length etc. The Ethernet uses bus topology.
- It uses a single coaxial cable. To which multiple computers connect. One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

ETHERNET SPEEDS: part of long and maybe short

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

ENCODING USED IN ETHERNET:

The encoding used in Ethernet is Manchester encoding. It uses signal changes to encode data. e.g. A change from positive voltage to 0 encodes.



Lecture No. 8 CARRIER SENSE MULTIPLE ACCESS (CSMA)

CARRIER SENSE MULTIPLE ACCESS (CSMA) vimp short

- There is no central control management when computers transmit on Ethernet. For this purpose the Ethernet employs CSMA to coordinate transmission among multiple attached computers.
- CSMA is a coordination scheme that defines how to take turns using a shared cable. A computer listen to the codes i.e. it senses the carrier.
- If the cable is idle it starts transmitting and if the cable is in use then it waits. If simultaneous transmission occurs, the frames interfere with each other and this phenomenon is called collision.

COLLISION DETECTION:

The signals from two computers will interfere with each other and the overlapping of frames is called a collision. It does not harm to the hardware but data from both frames is grabbed.

ETHERNET CD:

To detect the collision, Ethernet interfaces include hardware to detect transmission. It performs two operations:

- It monitors outgoing signals.
- Grabbled signal is interpreted as a collision.

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission.

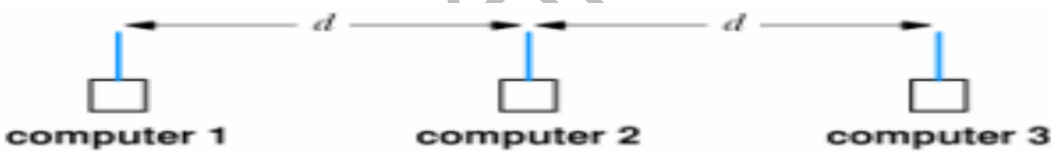
WIRELESS LANs AND CSMA/CA: vimp short question

IEEE 802.11 is standard wireless LAN that uses radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances. It uses shared medium and radio waves instead of coaxial cable.

LIMITED CONNECTIVITY WITH WIRELESS: vimp LONG question

In contrast with wired LANs, not all participants may be able to reach each other. Because:

- It has low signal strength.
- In wireless LANs the propagation is blocked by walls etc.
- It can't depend on CD to avoid interference because not all participants may hear.



CSMA/CA

- Wireless uses Collision Avoidance rather than Collision Detection
Transmitting computer sends very short message to receiver responds with short message reserving slot for transmitter.
- Response from receiver is broadcast, so all potential transmitters receive reservation.

Local Talk vimp short

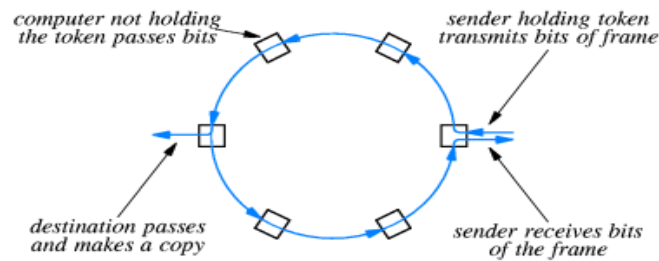
- LAN technology invented by Apple that uses bus topology
- Interface included with all Macintosh computers

- Relatively low speed - 230.4Kbps
- Low cost easy to install and connect
- Uses CSMA/CA

Token Ring

short/ Important Long

- Many LAN technologies that use ring topology use token passing for synchronized access to the ring.
- Ring itself is treated as a single, shared communication medium.
- Bits pass from transmitter, past other computers and are copied by destination.
- Hardware must be designed to pass token even if attached computer is powered down.



Using the token

part of long

- When a computer wants to transmit, it waits for the token.
- After transmission, computer transmits token on ring.
- Next computer ready to transmit receives token and then transmits.

Token and Synchronization

part of long

Token gives computer permission to send one frame If all ready to transmit, enforces ``Round-Robin" access If none ready to transmit, token circulates around ring.

IBM Token Ring

LONG

- Very widely used

- Originally 4mbps, now 16Mbps
- Uses special connector cable between computer and ring interface

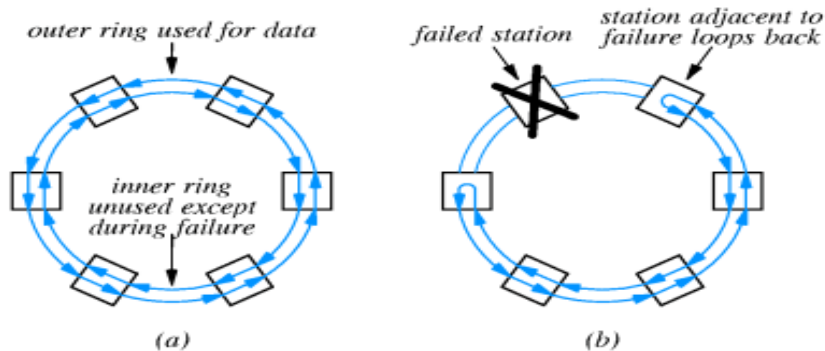
FDDI

Fiber Distributed Data Interconnect (FDDI) is another ring technology

- Uses fiber optics between stations
- Transmits data at 100Mbps
- Uses pairs of fibers to form two concentric rings

FDDI and Reliability

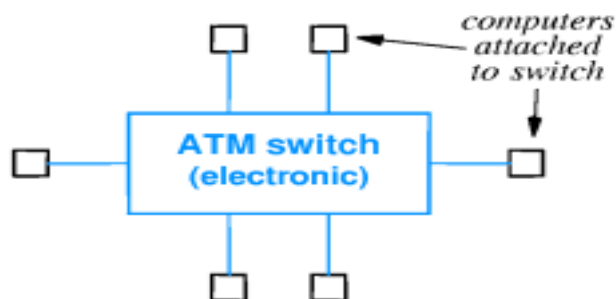
- ♦ FDDI uses counter-rotating rings in which data flows in opposite directions.
- ♦ In case of fiber or station failure, remaining stations loop back and reroute data through spare ring.
- ♦ All stations automatically configure loop back by monitoring data ring.



ATM – Star Network

LONG

- ♦ Asynchronous Transfer Mode technology consists of electronic packet switches to which computers can connect.
- ♦ ATM switches form hub into which computers connect in a star topology.
- ♦ Computers get point-to-point connections - data from transmitter is routed directly through hub switches to destination.
 - If none ready to transmit, token circulates around ring
 - If all ready to transmit, enforces ``Round-Robin" access



ATM Details

- ◆ Transmits data at over 100Mbps
- ◆ Uses fiber optics to connect computer to switch
- ◆ Each connection includes two fibers

Lecture No. 9 HARDWARE ADDRESSING

HARDWARE ADDRESSING

- Need to devise technique for delivering message through LAN medium to single, specific destination computer.
- Sending computer uses a Hardware Address to identify the intended destination of a frame.
- Sending computer also identifies Type of data carried in the frame.

Specifying a Destination

Data sent across a shared network reaches all attached stations - for all LAN topologies. Interface hardware detects delivery of frame and extracts frame from medium but most applications want data to be delivered to one specific application on another computer - not all computers.

LAN Interface

LONG

- LAN interface handles all details of frame transmission and reception
- Adds hardware addresses, error detection codes, etc. to outgoing frames
- May use DMA to copy frame data directly from main memory
- Obeys access rules when transmitting Checks error detection codes on incoming frames
- It May use DMA to copy data directly into main memory Checks destination address on incoming frames
- Station's address, a copy of the frame is passed to the attached computer. Frames not addressed to the local computer are ignored and don't affect the local computer in any way.

Format of Hardware Addresses

- ◆ Numeric value

- ♦ Size selected for specific network technology
- ♦ Length is one to six bytes

Assigning Hardware Addresses LONG

Hardware addresses must be unique on a LAN. There are three categories of address forms:

- Static
- Configurable
- Dynamic

STATIC:

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

CONFIGURABLE:

In this category, the address can be set by the end user either manually. The system administrators must coordinate to avoid the conflict.

DYNAMIC:

In this category the interface automatically assigns physical address each time it is powered up. This automatic scheme must be reliable to prevent conflicts.

BROADCASTING: short question

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations. A special broadcast address is used to identify broadcast message, which are captured by all stations.

Lecture No. 10 FRAME TYPE IDENTIFICATION

MULTICASTING:

Multicasting is defined as a single source sending to multiple recipients on a network when the receiver broadcasts a signal for acceptance. Multicasting has its own Class D IP

addressing scheme, controlled and assigned by the Internet Assigned Numbers Authority (IANA).

The interface hardware frame then begins accepting three types of frames:

- Multicast frames
- Broadcast frames
- The frames that are destined to the station itself.

Multicast Addressing

If an application program wishes to receive certain frames, then it programs the interface hardware to accept an additional set of addresses.

The interface hardware then begins accepting those frames (multicast frames), broadcast frames and frames that are destined to the station itself.

- ♦ For example: Computers running an audio application can receive audio frames, if the interfaces are programmed to receive them. The other computers that are not running that audio application will not waste resources.

Identifying Packet Contents OR Explicit Frame Type and Implicit Frame Type

LONG

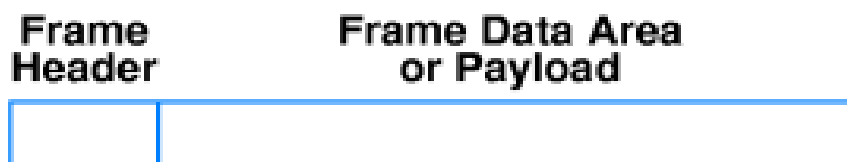
Destination must get some clue about how to interpret frame data.

Identifying Packet Contents Can use: Explicit Frame Type - Identifying value included with frame describes type of included data. Implicit Frame Type - Receiver must infer type from frame data.

Headers and Frame Formats

LAN technology standards define frame format for each technology.

All contemporary standards use the following general format: Frame header has address and other identifying information. Information typically in fields with fixed size and location. Data area may vary in size.



Frames without Type Fields

Some LAN technologies do not include a type field.

Sender and receiver can agree on interpretation: Agree on a single data format and use only that format. Limits LAN to one type of data. All computers on LAN must use one format. Agree to encode the data format in the first few bytes of the data field.

Encoding the Data Type

Illustration of using data area to encode data type: To ensure interoperability, format of encoding area must be universally agreed upon. Format typically set by standards body.



IEEE 802.2 LLC

- ♦ To ensure IEEE 802.2 standard includes Logical Link Control (LLC) Sub Network Attachment Point (SNAP) header.
- ♦ SNAP/LLC format widely used; e.g., by Ethernet.
- ♦ LLC portion indicates SNAP field to follow.
- ♦ OUI (Organizationally Unique Identifier) identifies Ethernet specification organization.
- ♦ TYPE field interpreted as in Ethernet (in this case, IP).



Unknown Types

For either encoding format, some computers may not be prepared to accept frames of some types like: Protocol type not installed. Newly defined type.

Receiving computer examines type field and discards any frames with unknown type.

Network Analyzers

LONG

A network analyzer or network monitor or Network Sniffer is used to examine the performance of or debug a network. Can report statistics such as capacity utilization, distribution of frame size, collision rate or token circulation time.

Operation of Network Analyzer

Basic idea is a computer with a network interface that receives all frames. Sometimes called Promiscuous Mode. Many desktop computers have interface that can be configured for promiscuous mode. Combined with software, computer can examine any frame on LAN. Communication across a LAN is not guaranteed to be private. Computer receives and displays (but does not respond to) frames on the LAN.

Filtering Incoming Frames

Analyzer can be configured to filter and process frames.

- Count frames of a specific type or size.
- Display only frames from or to specific computers.
- In general, can be configured to match value of any field and capture only those frames meeting the filter specification.

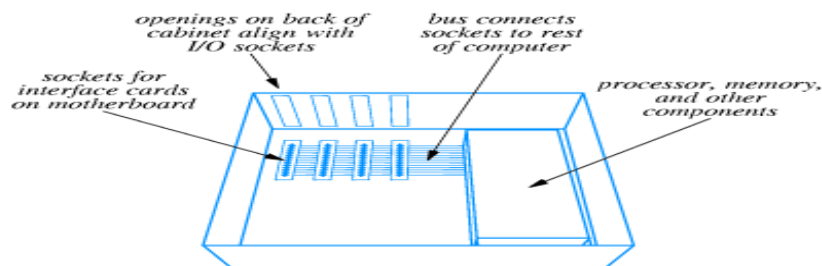
Lecture No. 11 INTERFACE HARDWARE

Speeds of LANs and Computers

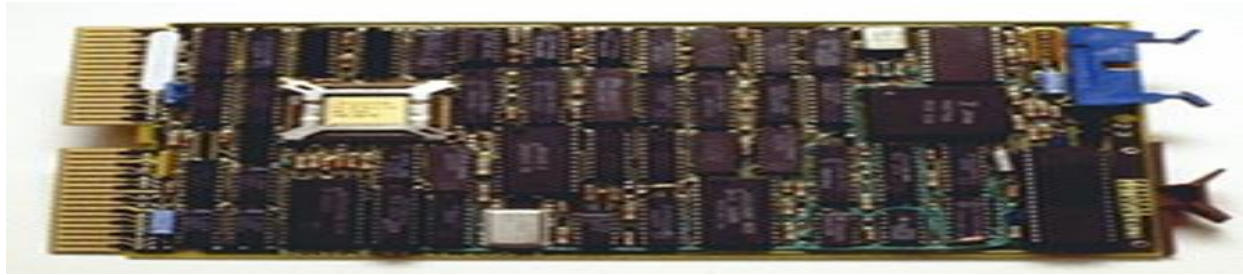
LAN data transmission speeds are typically fast relative to CPU speeds. LAN speeds are defined independent of any specific processor speeds. Allows for mix of attached systems. New computers can be attached without affecting LAN speeds.

Network Interface Hardware

CPU can't process data at network speeds. Computer systems use special purpose hardware for network connection typically a separate card in the backplane. Network adapter card or network interface card (NIC). Connector at back of computer then accepts cable to physical network.



Network Interface Card (NIC)



NICs and Network Hardware

NIC is built for one kind of physical network Ethernet interface can't be used with token ring ATM interface can't be used with FDDI Some NICs can be used with different, similar hardware Thick, thin and 10Base-T Ethernet 10Mbps and 100Mbps Ethernet.

NIC and CPU Processing

important LONG and short

NIC contains sufficient hardware to process data independent of system CPU. Some NICs contain separate microprocessor. Includes analog circuitry, interface to system bus, buffering and processing. Looks like any other I/O device to system CPU. System CPU forms message request Sends instructions to NIC to transmit data Receives interrupt on arrival of incoming data.

Connection between NIC and Physical Network

important LONG

Two alternatives:

NIC contains all circuitry and connects directly to network medium. Cable from NIC connects to additional circuitry that then attaches to the network medium. Thin Ethernet vs. 10Base-T both are Ethernet; network technology not limited to one style of connection.

Thick Ethernet wiring

important LONG and short

- Uses thick coax cable.
- AUI cable (or transceiver or drop cable connects from NIC to transceiver.
- AUI cable carries digital signal from NIC to transceiver.
- Transceiver generates analog signal on coax.
- Wires in AUI cable carry digital signals, power and other control signals.
- Thick Ethernet also requires Termination to avoid signal reflectance.

Connection Multiplexing

In some circumstances, transceivers may be inconvenient; e.g., workstations in a lab.

Connection Multiplexer connects multiple computers to a single transceiver each computer's AUI cable connects to connection multiplexor One AUI from multiplexor to Ethernet coax.

Thin Ethernet Wiring

important LONG and short

- Uses thin coax that is cheaper and easier to install than thick Ethernet coax.
- Transceiver electronics built into NIC; NIC connects directly to network medium.
- Coax cable uses BNC connector.
- Coax runs directly to back of each connected computer.
- T-Connector attaches directly to NIC.

Lecture No. 12 LAN WIRING AND PHYSICAL TOPOLOGY

10Base-T

Formally known as 10BaseT. Does not use coaxial cable, uses twisted pair wiring with RJ-45 connectors. Each computer has a dedicated connection to a central electronic device called an Ethernet Hub.

Hubs

Extension of connection multiplexing concept sometimes called “Ethernet-in-a-Box”. Electronic components in a Hub simulate a shared physical cable, making the entire system operate like a conventional Ethernet.



Advantages and Disadvantages of Wiring Schemes

LONG and Short

Each of the three wiring schemes has advantages and disadvantages:

- **Reliability issues:**

- Wiring that uses a transceiver for each connection does not affect the entire network if a transceiver cable is disconnected.
- A cable cut occurring in a Hub wiring only affects one computer.
- **Cost issues:**
 - Twisted pair Ethernet is the cheapest wiring that makes it so popular.
 - Thicknet is the most costly wiring which is no longer used
- ♦ **Example**
 - Eight offices wired with a) Thick Ethernet, b) Thin Ethernet, c) Twisted Pair Ethernet

The Topology Paradox

Question:

Is Ethernet a bus topology, or does the topology depend on the wiring?

Answer:

Both. Twisted pair Ethernet forms a classic star topology, however functions like a bus 10BaseT Ethernet is often called a star-shaped bus.

Two different types of topologies:

- Logical Topology: defined by the specific network technology
- Physical Topology: depends on the wiring scheme

10/100 NETWORK INTERFACES AND AUTONEGATIATION:

LONG and Short

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T Twisted pair Ethernet operates at 100Mbps.

100Base-T technology is backward compatible and allows the participants to negotiate a speed when connection is established. This process is known as auto negotiation.

Categories of Wires

LONG and Short

Cable used for wiring should match:

- The intended data rate
- The distance between the devices
- The amount of electromagnetic noise
- Anticipated future needs

➤ COST

Category	Bandwidth	Typical Uses
3	16 MHz	older, low-speed networks; analog telephones
4	20 MHz	short distance 10Base-T
5	100 MHz	10Base-T Ethernet; some 100Base-T
5E	100 MHz	100Base-T (Fast Ethernet); some 1000Base-T
6	250 MHz	1000Base-T Gigabit Ethernet) or ATM
7	600 MHz	future (possibly 10 Gigabit Ethernet)

Lecture No. 13 FIBER MODEMS AND REPEATERS

Distance Limitation And LAN Design

The maximum cable length of a LAN is fixed, because The electrical signal level gets weaker as it travels The delays must be short to allow access mechanisms (CSMA/CD, token passing), work properly.

LAN Extensions

- Several techniques extend diameter of LAN medium.
- Most techniques use additional hardware.
- LAN signals relayed between LAN segments.
- Resulting mixed technology stays within original engineering constraints while spanning greater distance.

Fiber Optic Extensions

LONG And short

The Fiber Modems Convert digital data into pulses of light Transmit over the optical fiber Receive light and convert into digital data Because Delays on optical fiber is low and Bandwidth is high This mechanism will successfully extend the LAN across several kilometers.

Repeaters

LONG And short

- Fact: Electrical signals gets weaker while travelling over copper.

- A repeater (a hardware device) Amplify the weakening signals received from one segment And retransmit them on to another segment.
- One repeater doubles, two repeaters triples the maximum cable length limitation.
- Computers attached to different segments communicate as if they are connected to the same cable.

Question: Can we increase the maximum cable length as many times as we wish by just adding repeaters?

Answer: No Every repeater introduces a delay Access mechanisms such as CSMA/CD does not work with long delays Ethernet standard specifies that any two stations cannot be separated by more than four repeaters.

Disadvantages:

Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signals.If a collision or error occurs in one segment, Repeaters amplify and retransmits also the error onto the other segments.

Bridges

short

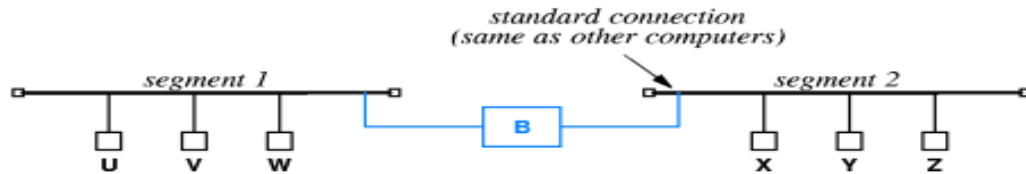
A Bridge is a hardware device also used to connect two LAN segments to extend a LAN Unlike a repeater, a bridge:

- Uses two NICs, to connect two segments
- Listens to the all traffic
- Recognizes frame format
- Forwards only correct, complete frames
- Discards the collided and errored frames

Frame Filtering

short

A typical bridge has two NICs, a CPU, a memory and a ROM.It only runs the code stored in its ROM.The most important task a bridge performs is frame filtering.If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if the frame is destined to the that segment.Broadcast and multicast frames are also forwarded.



Event	Segment 1 List	Segment 2 List
Bridge boots	—	—
U sends to V	U	—
V sends to U	U, V	—
Z broadcasts	U, V	Z
Y sends to V	U, V	Z, Y
Y sends to X	U, V	Z, Y
X sends to W	U, V	Z, Y, X
W sends to Z	U, V, W	Z, Y, X

Lecture No. 14 BRIDGES

STARTUP AND STEADY STATE:

When a bridge first boots, the address lists are empty (in startup state). The bridge forwards frames to the other segment if it can not find its destination address in its lists. After sometime, when the bridge has received at least one frame from every computer, it has the lists built (steady state), and it forwards frames as far as it is necessary.

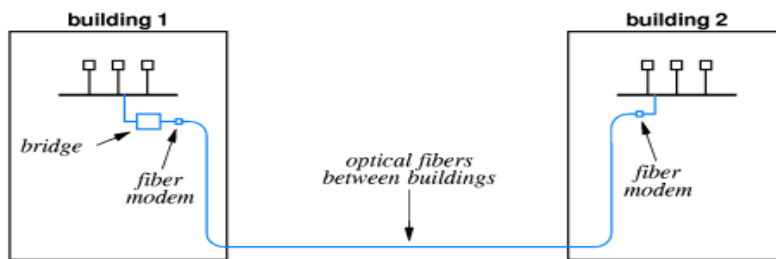
Planning a Bridged Network

In a steady state, a bridge allows simultaneous use of each segment. When designing a LAN, bridges can be installed to divide the LAN into segments to improve performance.

For example: Frequently contacting computers can be attached to the same segment. The frame traffic on one segment does not affect the other segment.

Bridging Between Buildings

If two buildings are located far from each other, a bridge, a pair of fiber modems and an optical fiber can be used to connect two LANs.



Advantages:

- Cost: An optical fiber and fiber modems are sufficient to connect many computers located on separate buildings
- Maintenance: No need to change the wiring between the buildings when installing and removing a computer
- Performance: The traffic on each building does not affect the other

Bridging Across Longer Distances

There are two common methods to connect two distant sites:

- Leased serial line connection Less expensive
- Leased satellite channel Can span arbitrarily long distance

Unlike optical fibers, satellite connections are low-bandwidth to save cost. Because the frames arrive from local network much faster than they can be sent across a satellite link,

- The bridge must use buffering (saving a copy of frame into memory until it can be sent)
- The bridge may run out of memory
- The communication software usually waits for a response after sending a few frames

A Cycle of Bridges

A bridged network can connect many segments. One bridge is needed to connect each segment to the rest of the bridged network. If the bridges on a large network form a cycle, then broadcasting frames introduce a problem. Copies of a broadcast frame continuously flow around the cycle, each computer receiving an infinite number of copies.

Lecture No. 15 SWITCHES AND WAN TECHNOLOGIES

Switching **short**

A switched LAN consists of a single electronic device that transfers frames among the connected computers. A hub with multiple ports simulates a single shared medium. However, a switch simulates a bridged LAN with one computer per segment. If a hub is used to connect many computers on a LAN, then only two computers can communicate at a given time. However, if a switch is used, the communication between two computers will not affect the others, pairs of computers can communicate at the same time.

Combining Switches and Hubs

To reduce costs, computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch.

- Hubs simulate single shared LAN segments
- Switch simulates a bridged LAN connecting segments

BRIDGING AND SWITCHING WITH OTHER TECHNOLOGIES:

Hubs, bridges and switches are not limited to Ethernet logical bus topology. They are available also for other networking technologies such as Token ring, FDDI etc. Like FDDI hub or Token ring hub.

WAN TECHNOLOGIES AND ROUTING: **LONG**

LANs can be extended using techniques in previous chapter. They cannot be extended arbitrarily for or to handle many computers. Because there are distance limitations even with extensions so we need other technologies for larger networks.

CHARACTERIZATION OF NETWORKS: **LONG**

There are three types of characterization of networks.

- LOCAL AREA NETWORK (LAN): It is used for a single building.
- METROPOLITAN AREA NETWORK (MAN): It is used for a single city.
- WIDE AREA NETWORK (WAN): It is used for a country level networking and even for continents.

DIFFERENCE BETWEEN LAN AND WAN:

A LAN, abbreviated from Local Area Network, is a network that covers a small geographical area such as homes, offices, and groups of buildings. Whereas a WAN, abbreviated from Wide Area Network, is a network that covers larger geographical areas that can span the globe.

Packet Switches

Packet switching is **the transfer of small pieces of data across various networks**. These data chunks or “packets” allow for faster, more efficient data transfer. Often, when a user sends a file across a network, it gets transferred in smaller data packets, not in one piece.

Connections to Packet Switches

Packet switches may connect to computers and to other packet switches. Typically high speed connections to other packet switches, lower speed to computers. Technology details depend on desired speed.

Packet Switches as Building Blocks

Packet switches can be linked together to form Wans. Wans need not be symmetric or have regular connections. Each switch may connect to one or more other switches and one or more computers.

PHYSICAL ADDRESSING IN A WAN

The physical addressing in a WAN is similar as in LAN in the following way:

- The data is transmitted in packets equivalent to frames.
- Each packet has a format with header.
- The packet header includes destination and source addresses.
- Many WANs use hierarchical addressing for efficiency.

One part of address identifies destination switch. Other part of address identifies port on switch.

NEXT HOP FORWARDING:

Packet switch must choose outgoing connection for forwarding the packet. There are two cases.

- If the destination is local computer, packet switch delivers computer port. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
- The choice of another switch is based on destination address in packet.

Lecture No. 16 ROUTING

Source Independence

Next hop to destination does not depend on source of packet, called **Source Independence**. Allows fast, efficient routing. Packet switch need not have complete information, just next hop

Reduces total information Increases dynamic robustness - network can continue to function even if topology changes without notifying entire network.

Hierarchical Addressing and Routing

The Process of forwarding the packets from one hop to the next is known as Routing. In particular, all destinations on same switch have same next hop Thus, routing table can be collapsed:

Destination	Next Hop
(1, anything)	interface 1
(3, anything)	interface 4
(2, anything)	local computer

Routing in a WAN

More computers == more traffic. Can add capacity to WAN by adding more links and packet switches Packet switches need not have computers attached Interior switch - No attached computers Exterior switch - Attached computers.

Both Interior and Exterior switches:

- Forward Packets
- Need Routing Tables
- Must have:
- Universal Routing - Next hop for each possible destination
- Optimal Routes - Next hop in table must be on shortest path to destination

Use a graph:

- Nodes model switches
- Edges model direct connections between switches
- Captures essence of network, ignoring attached computers

destin- ation	next hop	destin- ation	next hop	destin- ation	next hop	destin- ation	next hop
1	-	1	(2,3)	1	(3,1)	1	(4,3)
2	(1,3)	2	-	2	(3,2)	2	(4,2)
3	(1,3)	3	(2,3)	3	-	3	(4,3)
4	(1,3)	4	(2,4)	4	(3,4)	4	-
<i>node 1</i>		<i>node 2</i>		<i>node 3</i>		<i>node 4</i>	

Lecture No. 17 ROUTING ALGORITHMS

BUILDING ROUTING TABLES: Short Question

There are basically two methods for building routing tables, which are as follows:

- Manual entry
- Software

Further there are two methods for computing routing table information.

- Static routing
- Dynamic routing

- **STATIC ROUTING:**

It is done at boot time. It is simple and has low network overhead. It is inflexible.

- **DYNAMIC ROUTING:**

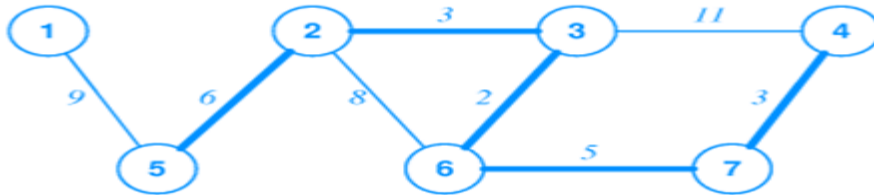
It allows automatic updates by a programmer. It can work around network failures automatically.

Computing Shortest Path in a Graph

Assume graph representation of network at each node. Use Dijkstra's algorithm to compute shortest path from each node to every other node. Extract next-hop information from resulting path information. Insert next-hop information into routing tables.

Weighted Graph

Dijkstra's algorithm can accommodate weights on edges in graph. Shortest path is then the path with lowest total weight (sum of weights of all edges). Shortest path not necessarily fewest edges (or hops).



DISTANCE-VECTOR ROUTING:

Local information is next hop routing table and distance from each switch. The switches periodically broadcast topology information i.e. destination, distance. Other switches update routing table based on received information.

Diff between DISTANCE-VECTOR ROUTING and LINK-STATE ALGORITHM

Short/LONG Question

DISTANCE-VECTOR ROUTING:

- It is very simple to implement.
- Packet switch updates its own routing table first.
- It is used in RIP.

LINK-STATE ALGORITHM:

- It is much more complex.
- Switches perform independent computations.
- It is used in OSPF.

FRAME RELAY:

It is used for Telco service for delivering blocks of data. It is connection based service and must contract with Telco for circuit between two endpoints. It is typically 56kbps or 1.5Mbps and can run to 100Mbps.

SMDS:

Switched Multi-megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It typically ranges from 1.5Mbps to 1000Mbps.

ATM (ASYNCHRONOUS TRANSFER MODE):

It was designed as a single technology for voice, video and data and has low jitter (variance in delivery time) and high capacity. It uses fixed size, small cells, 48 octet's data and 5 octets header. It can also connect multiple ATM switches into a network

Lecture No. 18 CONNECTION-ORIENTED NETWORKING AND ATM

CONNECTION-ORIENTED NETWORKING AND ATM

Short Question

LANs and WANs can both connect multiple computers, but they have different base technologies and meet different goals. ATM is a single technology that is designed to meet the goals of both LANs and WANs. ATM uses the concept of connection-oriented networking.

JITTER:

Jitter is the term used for variance in transmission delays. Jitter is significance for voice, video and data. In LANs, jitter can occur when a packet is delayed because the network is busy.

Packet Sizes

Short Question

Large packets result in less overhead because a smaller fraction of the packet is used for header information. Optimum networks use 4KB packets or larger.

Large packets can't easily be used for voice 8 bit samples (at 125 usec per sample) would require half a second to fill a 4KB packet. Echo cancellation can only be used with low transmission delays.

ATM Cells

To meet its goals, ATM uses small, fixed-sized packets called cells. Each cell has 53 octets. VCI/VPI fields identify the cell's destination. PRIO tell if cell can be discarded. To meet its goals, ATM uses small, fixed-sized packets called cells. Each cell has 53 octets. VCI/VPI fields identify the cell's destination. PRIO tell if cell can be discarded.

VIRTUAL CHANNEL (OR CIRCUITS):

LONG/ Short Question

The VC is identified by a 24-bit value formed from:

- The VPI, or virtual path indicator (8 bits), which identifies a particular path through the network;
&

- The VCI, or virtual channel indicator (16 bits), which identifies the channel in the virtual path is being used by the connection
- Most frequently, the 24-bit pair is treated as just a single connection identifier by computers

Lecture No. 19 ATM: VIRTUAL CIRCUITS

Labels and Label Switching

Short Question

An ATM network is built from interconnected ATM switches. Attachment points on switches, or ports, can be connected to computers or other ATM switches. As cells arrive at an ATM switch, their VPI/VCI is modified using a forwarding table which gives the new VPI/VCI for the next leg of the cell's trip. The forwarding table is essentially indexed by the incoming cell's VPI/VCI, and the contents yield the new VPI/VCI.

LABEL REWRITING:

The replacement of the incoming cell's VPI/VCI with a probably different VPI/VCI is called rewriting. ATM is thus called a label rewriting or label switching system. Thus two computers with a connection through an ATM network will likely have different VPI/VCI values for each end of the connection.

Permanent Virtual Circuits

Provisioning requires two steps:

- Determine a complete path (that is, identify the switches that will be used)
- Choose appropriate VCI/VPI for each step in the path, and configure each adjacent pair of switches (easy, since each switch rewrites the VCI/VPI)

ESTABLISHING AN SVC:

LONG/ Short Question

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready. If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established.

SIGNALING:

Short Question

In telephony, signaling is the exchange of information between involved points in the network that sets up, controls, and terminates each telephone call. In in-band signaling, the signaling is on the same channel as the telephone call.

ATM QoS SPECIFICATIONS:

LONG/ Short Question

There are three types of QoS specifications, which are given as follows:

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

AVAILABLE BIT RATE (ABR):

It is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

Lecture No. 20 ATM AND NETWORK OWNERSHIP

Cells vs Packets

ATM designers chose cells over packets because:

- Cells aren't variable length, and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells, bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable, and requires complicated interrupt schemes to detect completion of transmission.
- QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

NETWORK OWNERSHIP:

There are two categories in this case:

Private Network

Owned by single organization or company

Public Network

Owned by common carrier e.g. Phone Company)

ADVANTAGES AND DISADVANTAGES:

The advantages and disadvantages of Public and Private Networks are given in the table below:

NETWORKS	ADVANTAGES	DISADVANTAGES
PUBLIC	They are flexible.	There are no decision-making equipment or policies.
PRIVATE	The owner has complete control over both the technical decision and policies.	They are expensive to install and maintain.

Lecture No. 21 NETWORK SERVICE PARADIGM

VIRTUAL PRIVATE NETWORK:

Virtual Private Network (VPN) combines the features of both private and public networks. It is limited to single organization and uses public network for connectivity. These connections are sometimes called tunnels and connect sites. Each site sees tunnel as point-to-point link. There is no access for other users of public networks.

SIMPLEX & FULL DUPLEX CONNECTION:

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex design a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

CONNECTION DURATION AND PERSISTENCE:

Connection can be made on demand set up permanently. There are two types:

- Switched Connection or Switched Virtual Circuit (SVC).
- Permanent Connection or Permanent Virtual Circuit (PVC).

Permanent connections are originally hardwired and now configured at system unit time. In switched connections, computer maintains permanent connection to network and networks make connections on demand. Internal components are switched networks is a switched data network.

ADDRESSES AND CONNECTION IDENTIFIERS:

Address is a complete unique identifier. Connectionless delivery requires address on each packet. Connection-oriented delivery can use a short hand that identifies the connection rather than the destination.

Lecture No. 22 NETWORK PERFORMANCE

NETWORK PERFORMANCE

There are two types of characteristics in case of network performance.

- Delay
- Throughput

Delay

Important quantitative property of networks Delay is a measure of how long it takes for a bit of data to travel across the network from one computer to the other. Measured in seconds or fractions of seconds (milliseconds) Maximum and Average Delay.

Types of Delay

- ♦ Propagation delay - time to travel across medium.
- ♦ Switching delay - time for network. Component (hub, bridge, packet switch) to forward data.
- ♦ Access delay - time to get control of medium (CSMA/CD, token).
- ♦ Queuing delay - time enquired in packet switches.

THROUGHPUT:

Throughput is a measure of the rate at which data can be sent through the network. The throughput capability of the underlying hardware is called bandwidth. Because each frame contains headers, the effective throughput is less than the hardware bandwidth. Networking professional often use the term speed as a synonym for throughput.

RELATIONSHIP BETWEEN DELAY AND THROUGHPUT:

If a packet switch has a queue of packets waiting when a new packet arrives. The new packet will be placed on the entire queue and will need to wait while the switch forwards the previous packets. Throughput and delay are not completely independent. As traffic in a computer network increase, delays increase a network that operates at close to 100% of its throughput capacity experiences severe delay.

JITTER:

The amount of delay that a network introduces is called jitter. A network with zero jitter takes exactly the same amount of time to transfer each packet. A network with high jitter takes much longer to deliver some packets than others.

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